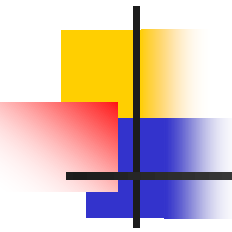
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Computer Programming II (CSE2035)

Lab #9: STL

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Dept. of Computer Science and Engineering
Sogang University

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Homework 9: Lecture Revisited (15pt)

- In this homework, we will reimplement our previous Lecture and Student classes from Homework 6
- Let's rewrite these classes in *C++ style*
 - Using STL and C++ string

Part 1: Rewriting Lecture Class

- First, let's rewrite the class for managing lecture information
- Properties (member variables) are defined in `lecture.hpp` file
- This time, we will use `std::string` and `std::map`

```
enum Grade { A, B, C, NA };
...
class Lecture {
private:
    std::string code;
    std::map<int, Grade> id_to_grade;
public:
    ...
}
```

(Mapping from student no. to grade)

Student No.	Grade
101	A
205	C
404	B

Part 1: Rewriting Lecture Class

- **Several functions (constructor, destructor, ...) are also defined in the header file (do not fix this file)**
 - Q) Is it okay to do nothing in the destructor? We have `std::string` and `std::map` objects as properties
 - A) Yes. When you include another object as a property, its destructor is automatically executed

```
class Lecture {
private:
    std::string code;
    std::map<int, Grade> id_to_grade;
public:
    Lecture(const char * _code) {
        code = _code; // Automatically converted to std::string
    }
    ~Lecture() { } // You don't have to do anything
    ...
}
```

Part 1: Rewriting Lecture Class

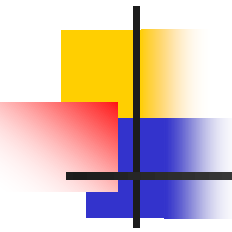
- **Implement the following functions in `lecture.cpp`**
 - Detailed specifications are given in the comment of `lecture.hpp` header file, so read them carefully
 - For example, you cannot set the grade of student who has not registered yet

```
bool AddStudent(int id);  
  
bool SetGradeOf(int id, Grade grade);  
  
bool GetGradeOf(int id, Grade *pGrade);  
  
int CountGrade(Grade grade); // Newly added!
```

Part 2: Rewriting Student Class

- Next, let's rewrite the class that can represent a student
- Properties (member variables) and several functions are defined in `student.hpp` file (**do not fix this file**)
- This time, we will use `std::vector` to store the Lecture object pointers

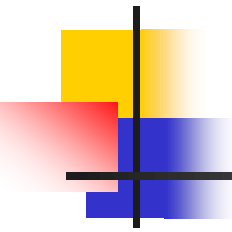
```
class Student: public UnivMem {  
private:  
    std::vector<Lecture*> lectures;  
public:  
    Student(int _id, const char* _name) { ... }  
    ...  
}
```



Part 2: Rewriting Student Class

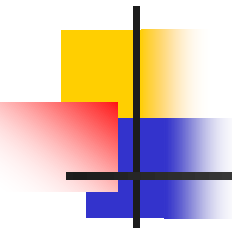
- **Implement the following functions in `student.cpp`**
 - Detailed specifications are given in the comment of `student.hpp` header file, so read them carefully

```
void PrintGrade();  
  
bool Register(Lecture *lec);
```

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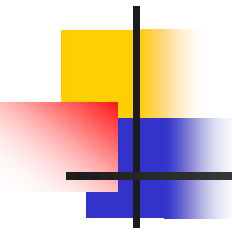
Note

- Not a difficult assignment, once you get familiar with using STL containers
- If you don't know what to do, check the example code uploaded in the cyber campus
- Also, you may have to check the reference site
 - Ex) <https://en.cppreference.com/w/cpp/container/map>
- After finishing the assignment, compare it with your previous solutions to hw6-1 and hw6-2
 - We can manage the data more conveniently

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Submission Guideline

- **Submit the completed `lecture.cpp` and `student.cpp` files to “Lab #9 Assignment” in cyber campus**
 - You can fix `main.cpp` to test your code, but don't submit it
 - Do not zip the directory, just upload these files directly
 - **Please do not change the file names**
 - If the submitted file doesn't compile with the "make" command, cannot give you any point
- **Deadline is 12/2 (Fri) 23:59**
- **Late submission is allowed until 12/5 (Mon) 23:59**
 - 20% deduction
- **Resubmission will be allowed until 12/9 (Fri) 23:59**
 - 40% deduction

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Q & A Session
